(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JUNE 2024

Paper Code: ES-108 ***Subject: Electrical Science***

Time: 3 Hours ***Maximum Marks: 60***

Note: Attempt five questions in all including question no.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1 Attempt **all** **(4x5=20)**

(a) An alternating current is given by i = 14.14 sin 377 t. Find the time taken for the current to reach 10 Ampere for the first time after passing through zero value.

(b) What are the different losses in transformers?

(c) What is the meaning of time domain analysis of first order R-L and R-C circuits?

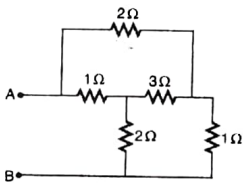
(d) What are the two difference between induction and synchronous motor?

(e) What are the different torques associated with measuring instruments?

UNIT-I

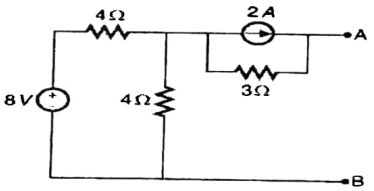
Q2 (a) Find the equivalent resistance R_{AB} between terminal A & B in Figure1 by Star-Delta transformation. **(6)**

Figure-1



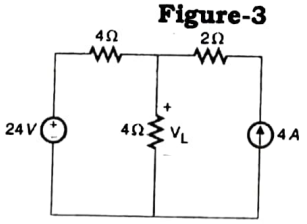
(b) Find Thevenin's equivalent of the circuit shown in Figure 2. **(4)**

Figure-2



Q3 (a) Find the voltage V_L in the circuit shown in Figure 3 using the principle of superposition. **(3)**

Figure-3



(b) State and prove maximum power transfer theorem. **(7)**

UNIT-II

Q4 (a) An inductor coil takes 10 A and dissipates1000 W when fed from 250 V 25Hz supply. Find the impedance, resistance, inductance, and power factor of the coil. **(6)**

(b) Explain the series resonance phenomenon in the series R-L-C circuit and derive the bandwidth formula of the circuit. **(4)**

P.T.O.